

FROM SYMBOLS TO MEGAYEARS: GLOBULAR-CLUSTER NUMBERS FOR THE CONE'S GRAVITATIONAL BATH

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ABSTRACT. [1] derives a diffusion coefficient $\text{Tr } D$ and friction coefficient A_{friction} for a charged particle in a plasma from binary Rutherford encounters, and applies them only to that idealized model and, separately in [2], to real fusion-plasma numbers. The gravitational bath in [1] was left at the level of a toy constant-diffusion model. This note completes it the same way: we first check, by direct Monte Carlo, that the plasma coefficients transfer unmodified to attractive (gravitational) binary encounters with $\kappa_{\text{int}} = Gm_t m_f$; then we apply them to typical globular-cluster-core parameters ($n = 10^4 \text{ pc}^{-3}$, $m = 0.5 M_\odot$, one-dimensional velocity dispersion $\sigma = 5 \text{ km/s}$, core radius 1 pc) and compute a relaxation time of 12.1 Myr entirely from first principles. Comparing against the standard, properly cited two-body relaxation formula $t_R \approx 0.34 \sigma^3 / (G^2 m^2 n \ln \Lambda)$ (Chandrasekhar 1942; Spitzer 1987; Binney & Tremaine 2008) gives $t_R \approx 90\text{--}104 \text{ Myr}$, a factor of ~ 8.5 larger. As with the plasma case, this traces substantially to the choice of characteristic velocity: using the definitionally consistent 3D rms speed $\sqrt{3}\sigma$ in place of σ itself closes most of the gap (residual factor 1.6), leaving a further, honestly unexplained factor we attribute to the same Maxwellian-averaging effects flagged for the plasma case. We separately note, without endorsing it as principled, that $u = 2\sigma$ matches the literature value to 6.4% — a striking but currently unexplained coincidence we report rather than quietly adopt.

1. DOES THE FORMULA TRANSFER TO GRAVITY?

The formulas of [1], Section 4, were derived and verified for like-charge (repulsive) Coulomb scattering. Gravity is always attractive; before applying anything, we check whether this matters.

Proposition 1.1 (Transfer to attractive encounters). *The Rutherford deflection formula $\theta = 2 \arctan(b_{90}/b)$ depends on the coupling only through $|\kappa_{\text{int}}|$, not its sign; consequently $\text{Tr } D$ (eq. 4 of [1]) and A_{friction} (eq. 5) hold unmodified for attractive encounters with $\kappa_{\text{int}} = Gm_t m_f > 0$.*

Numerical Verification. Repeating the binary-encounter Monte Carlo of [1] with an attractive force ($\kappa_{\text{int}} = Gm_t m_f$, 3×10^6 trial windows) gives $\text{Tr } D_{\text{MC}} / \text{Tr } D_{\text{analytic}} = 0.848$ and $A_{\text{friction,MC}} / A_{\text{friction,analytic}} = 0.911$ — the same $\sim 10\text{--}15\%$ leading-log accuracy as the repulsive case, confirming the formulas transfer with no sign-dependent correction.

2. A REAL SYSTEM: A TYPICAL GLOBULAR-CLUSTER CORE

Textbook-typical (not fit to any named cluster) core parameters:

$$n = 10^4 \text{ pc}^{-3}, \quad m = 0.5 M_\odot, \quad \sigma = 5 \text{ km/s (1D)}, \quad R = 1 \text{ pc}.$$

Our Own Prediction. Using only [1]’s formulas, $u = \sigma$ initially, and $\ln \Lambda = \ln(R/b_{90})$ computed from first principles (no screening length exists for gravity, so the natural cutoff is the system size):

$b_{90} = \kappa_{\text{int}}/(\mu_{\text{red}}\sigma^2)$	$5.31 \times 10^{12} \text{ m}$
$\ln \Lambda = \ln(R/b_{90})$	8.67
$\text{Tr } D$	$6.53 \times 10^{-8} \text{ m}^2/\text{s}^3$
$\tau = \sigma^2/\text{Tr } D$	12.1 Myr

3. COMPARISON TO ESTABLISHED STELLAR DYNAMICS

Proposition 3.1 (The standard two-body relaxation time). [3, 4, 5], *as summarized in [6]:*

$$(9) \quad t_R \approx \frac{0.34 \sigma^3}{G^2 m^2 n \ln \Lambda}, \quad \sigma := \sqrt{\langle v^2 \rangle}/3 \text{ (one-dimensional dispersion).}$$

Numerical Evaluation. Using our own $\ln \Lambda = 8.67$: $t_R = 104 \text{ Myr}$, roughly $8.5\times$ our 12.1 Myr.

Proposition 3.2 (Chasing the gap). *Our τ scales as u^3 . The 1D-to-3D-rms convention $u = \sqrt{3}\sigma$ (the natural reading of (9)’s own definition of σ) gives $\tau = 55.9 \text{ Myr}$ against $t_R = 92.0 \text{ Myr}$ at the corresponding $\ln \Lambda = 9.77$: a residual factor of 1.6.*

Remark (What we do and do not explain). Factor 1.6 is close to, but we do not claim identical to, the $1.5\times$ residual left unexplained for Braginskii’s plasma formula in [2], and we attribute it to the same cause: (9) is derived by properly averaging the diffusion coefficients over the full (Maxwellian) relative-velocity distribution of test and field stars, not by evaluating a single-speed formula at one characteristic velocity. We have not carried out that averaging. Separately, and only as an observation: setting $u = 2\sigma$ (with no derivation offered) gives $\tau = 83.6 \text{ Myr}$ against $t_R = 89.3 \text{ Myr}$ at that $\ln \Lambda = 10.05$ — a 6.4% match, better than the principled $\sqrt{3}\sigma$ choice. We report this because it is true of the numbers, not because we can justify 2σ over $\sqrt{3}\sigma$ or $\sqrt{6}\sigma$ (the relative-speed rms for two independent dispersion- σ populations, which *overshoots* the literature value). Presenting an unexplained best-fit alongside the principled-but-worse comparison is the honest way to report this, rather than silently choosing whichever velocity makes the numbers agree.

4. DISCUSSION

The pattern from [2] repeats: formulas derived with zero reference to real astrophysical data, checked only against a purpose-built Monte Carlo simulation of an idealized encounter model, produce a relaxation time within an order of magnitude of a named, independently-derived textbook result for an unremarkable real system, with a substantial part of the gap traced to an identifiable and defensible convention difference rather than left as an unexplained fudge factor. The match here is less clean than the plasma case’s 5.4% — the principled choice leaves a $1.6\times$ residual rather than closing to leading-log accuracy — and we consider that an honest data point rather than something to explain away by reaching for the velocity convention that happens to fit best. Two real gaps remain unaddressed: neither this note nor [1] performs the actual Maxwellian average that both the Spitzer-type plasma formula and the Chandrasekhar-type stellar-dynamics formula (9) build in from the start, and neither engages with the well-known fact that (9) itself describes only local two-body relaxation, not the global evolution of a real cluster, for which orbit-averaging and the cluster’s density profile matter and which this note does not model.

AI ASSISTANCE DISCLOSURE

Large language model tools (Anthropic’s Claude) were used in the derivation, literature search, numerical computation, and L^AT_EX preparation of this note. This includes: the Monte Carlo verification of Proposition 2.1 (transfer to attractive encounters); a web search to locate a properly citable standard formula for the two-body relaxation time (Proposition 4.1); numerical evaluation of all derived and literature quantities; and the diagnostic velocity-convention calculation in Proposition 4.2, including the explicit decision to report the unexplained 2σ near-match alongside, rather than instead of, the principled $\sqrt{3}\sigma$ comparison. All numerical claims were independently verified by the author.

REFERENCES

- [1] J. Ebert, *The Cone in a Bath: Boson, Gravitational, and Plasma Relaxation of the Semi-Major Axis*, Preprint (2026).
- [2] J. Ebert, *From Symbols to Microseconds: Fusion-Plasma Numbers for the Cone’s Bath Theorems*, Preprint (2026).
- [3] S. Chandrasekhar, *Principles of Stellar Dynamics*, University of Chicago Press, 1942.
- [4] L. Spitzer, *Dynamical Evolution of Globular Clusters*, Princeton Univ. Press, 1987.
- [5] J. Binney, S. Tremaine, *Galactic Dynamics*, 2nd ed., Princeton Univ. Press, 2008.
- [6] Diffusion and mixing in globular clusters, *arXiv:1801.01123*, Section 3.1 (statement of the two-body relaxation time formula and $\alpha = 9/(16\sqrt{\pi})$).

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